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Ultra-broadband and low-loss 3 dB optical power splitter based on adiabatic tapered silicon waveguides

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A broadband, low-loss and polarization-insensitive 3 dB optical power splitter based on adiabatic tapered silicon waveguides is proposed and investigated. 3D-FDTD simulation results show that the splitter achieves an output transmission efficiency of nearly 50% over an ultra-broad wavelength range from 1200 to 1700 nm. The device is fabricated, and experimental results show that the splitter exhibits a low excess loss of <0.19 dB for the TE polarization and <0.14 dB for the TM polarization over the entire measured wavelength range from 1530 to 1600 nm, while having an adiabatic taper length of only 5 μm . In addition, the measured power uniformity of the cascaded 1×8 splitter is only 0.47 dB, and 0.17 dB for the TE and TM polarizations, respectively. With the advantages of low loss, broad bandwidth, and compact size, the proposed splitter is a promising element for large-scale silicon integrated photonic circuits. © 2016 Optical Society of America

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A 3 dB optical power splitter is an important optical device that has been widely used for light power distributions and the construction of more complex optical devices, such as optical switches, modulators, multiplexers, and optical phased arrays [1–4]. Based on the silicon integrated platform, power splitters are mainly realized through Y-branches, directional couplers (DCs), and multi-mode interference (MMI) couplers. However, these approaches are conventionally limited by a number of drawbacks, such as the relatively high mode mismatch loss for the Y-branch when the branching angle is not sufficiently small, or the high sensitivity to wavelength and polarization for conventional DC and MMI couplers [5–7]. In this Letter, to realize a splitter with properties of low loss, broad bandwidth, strong polarization independence, and compact size simultaneously, we propose a novel splitter structure by using adiabatic tapered silicon waveguides based on the popular thin SOI platform that has a 220 nm thick

silicon layer and a 2 μm thick buried oxide layer (BOX). Tapering structure has been widely explored in realizing power splitters that are based on either adiabatic or non-adiabatic mode evolutions [8–15]. Unfortunately, these previously proposed splitters required a considerably long taper length (hundreds of μm) [8–11] or a complex-to-fabricate structure [12] to realize the adiabatic evolution, or were non-adiabatic-based which also suffered from relatively high radiation loss, such as the antenna-coupled-type Y-branches proposed in [13,14]. In [15], a compact Y-junction splitter based on an optimized tapering design is presented; although it exhibited relatively low loss (0.28 dB) at 1550 nm, the high-transmission-efficiency (>48%) bandwidth is relatively limited (<80 nm). In contrast, based on a very short adiabatic tapering design of only 5 μm long, the proposed splitter theoretically achieves an extremely high efficiency of nearly 50% for each output port over an ultra-broad wavelength range from 1200 to 1700 nm, and it is polarization insensitive. Experimental measurement results show that low excess loss (EL) of <0.19 dB and <0.14 dB can be achieved over the entire measured wavelength range from 1530 to 1600 nm for the TE and TM polarizations, respectively. At 1550 nm, the excess loss for the two polarizations is estimated to be only 0.11 and 0.08 dB.

The structure of the proposed splitter is shown in Fig. 1(a), which mainly consists of three regions, namely input waveguide, tapers (one input taper and two output tapers), and output bending waveguides. A single-mode input strip waveguide with a dimension of $W \times 220$ nm is gradually tapered down (tip width T) along a length L , and the taper is symmetrically inserted into two output tapers which have the same length (L), end width (W), and tip width (T) as the input taper. The ends of the output tapers are connected with two bending waveguides. The gap between two adjacent tapers is denoted as G .

Different from a 2×2 3 dB adiabatic coupler [8–11], for which the adiabatic coupling occurs between two tapered waveguides, in the proposed splitter, light is adiabatically coupled from one input waveguide to two symmetric output waveguides through three tapers and, as long as the adiabatic mode evolution is satisfied for both TE and TM polarizations, the 3 dB power splitting ratio

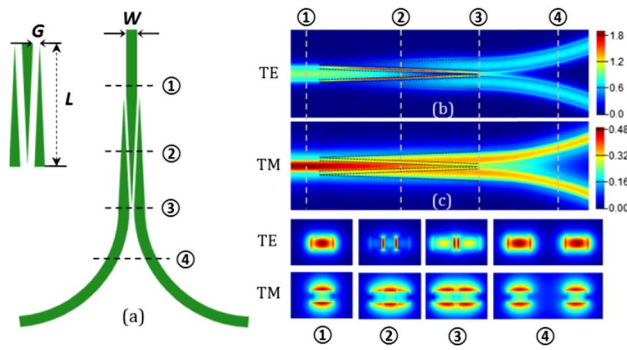


Fig. 1. (a) Structure of the proposed splitter and the electric field distributions along the device with (b) the TE and (c) the TM polarization input at the 1550 nm wavelength. The device parameters used in simulations are W (400 nm), L (5 μm), G (50 nm), and T (30 nm). In the simulation, the device is covered by 1.3 μm thick SiO_2 as the top cladding, and the refractive index of top SiO_2 was set to 1.46 based on the index measurement of PECVD-deposited SiO_2 by ellipsometer.

can be achieved for both polarizations at the end of the tapered region. The signal after passing through the tapers is then split into the fundamental modes through the bending waveguides. Figures 1(b) and 1(c) show the simulated mode evolution process for the TE and TM polarizations, respectively (in different color scales). It can be seen that when the TE mode is launched, due to the electric field discontinuities at the tapered region, the light field is first enhanced and concentrated within the tiny gaps and, along a certain propagation distance, light can be seen as lossless eigenmodes propagating in a double-slot waveguide [16]. On the other hand, for the TM mode, as the electric field is mainly concentrated at the top and bottom interfaces of the waveguide, the light field is not squeezed into the gaps throughout the tapers. Consequently, in practice, the mode transition in TE polarization is more sensitive to the sidewall roughness within the tapered region, which has to be taken into account in device optimization.

To achieve low loss and wide bandwidth for both TE and TM polarizations, the adiabatic mode evolution needs to be maintained for both polarizations so that only the fundamental mode is excited and propagates along the tapers [8–11]. An eigenmode solver was used for the mode analysis in the tapered region. The input and output waveguides width was first selected at 460 nm, and L was selected at 5 μm . As shown in Fig. 2(e), it is found that for either TE or TM polarization, two modes are supported along the tapers (denoted as the first- and second-order modes). Furthermore, the effective refractive index difference between these first- and second-order modes becomes smaller in the taper region along the light propagation direction. The effective indices at the taper end of the two modes in TE polarization are very close ($\Delta N_{\text{eff-TE}} = 0.05$), while the difference of the TM polarization is comparatively larger ($\Delta N_{\text{eff-TM}} = 0.2$). The corresponding mode profiles at the taper end cross section are shown in Figs. 2(a)–2(d). Therefore, in practice, when the TE-polarized signal is utilized, given the fabrication imperfection, the adiabatic transition condition is difficult to maintain and the higher-order mode can be excited, which can lead to mode interferences and cause extra losses [17]. Therefore, the device needs to be optimized to suppress the higher-order mode excitation, while still keeping the device compact.

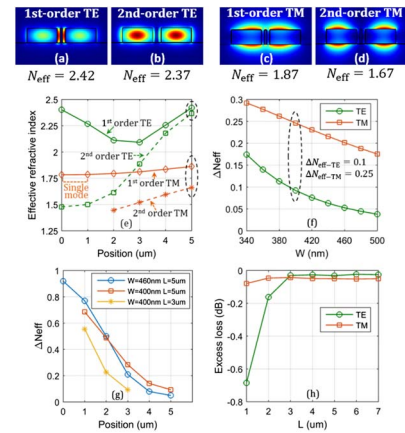


Fig. 2. (a)–(d) Mode profiles (1550 nm) simulated at the cross section of taper end with $W = 460$ nm. (e) Effective index of the four modes along the tapers with a length of 5 μm . (f) Simulated ΔN_{eff} at the taper end cross section with the change of W from 340 to 500 nm. (g) Change of ΔN_{eff} along the tapers for different W and L . (h) Simulated excess loss at 1550 nm with the change of L .

Theoretically, it is easier to maintain the adiabatic transition with a longer taper length (L), a smaller gap size (G), and a sharper tip width (T). However, considering fabrication limitations, G and T are chosen at 50 and 30 nm, respectively, based on the typical E-beam lithography (EBL) resolution. The impact of waveguide width (W) and taper length (L) is then investigated and optimized based on 3D-FDTD simulations. Figure 2(f) shows the simulated ΔN_{eff} at the taper end cross section when W is varied over a range from 340 to 500 nm. It can be seen that ΔN_{eff} decreases with W for both TE and TM polarizations, while overall it remains relatively large for the TM polarization. Since it is easier to maintain the adiabatic transition with a larger ΔN_{eff} , W is chosen at 400 nm here by taking the propagation and bending losses into consideration simultaneously. In addition to the larger ΔN_{eff} , another advantage of choosing $W = 400$ nm is that the change of ΔN_{eff} along the tapered region is also more smooth, as shown in Fig. 2(g).

With W fixed at 400 nm, the device performance in terms of EL at 1550 nm is then investigated by changing the taper length (L) from 1 to 7 μm , and the results are shown in Fig. 2(h). Note that in the simulation, the impact of sidewall roughness was not included in the device excess loss. In general, longer tapers are preferred to maintain the adiabatic transition. It is clear that for $L > 3$ μm , the EL for the TE polarization becomes negligible, indicating that no higher-order mode is excited, while for the TM polarization, due to the larger ΔN_{eff} , the EL remains low, even for $L < 3$ μm . Considering compact size, as well as the adiabatic requirements for both polarizations, a conservative value of 5 μm is chosen for L . As shown in Fig. 2(g), when $L = 5$ μm , ΔN_{eff} also changes more smoothly than the case when $L = 3$ μm . Compared with the previous polarization-independent 3 dB adiabatic couplers [8], the weak mode confinement at the input taper, as well as the small gap size between tapers, leads to much stronger coupling strength for both TE and TM polarizations, which leads to a significantly shorter (by two orders of magnitude) taper length which can satisfy the adiabatic evolution for both polarizations simultaneously.

Based on the optimized parameters, the simulated transmission spectra of the splitter are shown in Fig. 3(a). In the simulation, 90° bending waveguides were used as the output branches of the splitter, and the bending radius was set to a relatively large value of 15 μm to avoid substantial bending losses. As can be seen, the splitter achieves an extremely high and flat transmission efficiency of nearly 50% for each output port over the whole simulated wavelength range. For the TM polarization, due to the gradually increasing bending losses within the bending waveguide, the transmission efficiency gradually drops for the wavelengths >1550 nm. Nevertheless, the bandwidth with a high efficiency of >49% is still very broad (1200–1625 nm).

The splitter's tolerances to the variations of the nonzero taper tip width (T) and the small gap size (G) were also investigated. First, T was changed from 0 to 80 nm, while other parameters were kept at the optimized values. As shown in Fig. 3(a), the transmission efficiency degradation is negligible (<0.3%) for both polarizations. The impact on ΔN_{eff} is not substantial either, with $\Delta N_{\text{eff-TE}}$ dropping from 0.12 to 0.07 and $N_{\text{eff-TM}}$ dropping slightly from 0.25 to 0.23. Therefore, the splitter performance has relatively high tolerance to the taper tip width variation. Then G was increased from 50 nm to a relatively large value of 130 nm, while other parameters remained unchanged. As shown in Fig. 3(c), while the transmission ratio of the TE mode slightly decreases by around 0.6%, the transmission ratio of the TM mode remains stable. However, ΔN_{eff} reduces sharply for both polarizations, with $\Delta N_{\text{eff-TM}}$ dropping from 0.25 to 0.12 and $\Delta N_{\text{eff-TE}}$ dropping from 0.1 to only 0.02. As $\Delta N_{\text{eff-TM}}$ remains high (>0.12), even with large gaps, it is still difficult to excite the higher-order TM modes. For the TE polarization, as ΔN_{eff} rapidly drops to <0.05 when G is larger than 70 nm, the adiabatic evolution is difficult to maintain in practice. Therefore, the proposed splitter has higher fabrication tolerance in TM polarization, while relatively precise fabrication is needed for the TE polarization.

The proposed splitter was fabricated on an 8 inch SOI wafer with a 220 nm thick silicon layer and a 2 μm thick BOX layer. During device fabrication, EBL was used to define the structures with diluted ZEP (ZEP520A) as the positive-tone resist, and proximity effect correction (PEC) was applied to obtain the correct size of the small features in the tapered region. Cr was then deposited by E -beam evaporation to serve as the etching mask for silicon layer, and the silicon layer was etched by reactive-ion etching (RIE) with a SF₆/C₄F₈ chemistry. After removing Cr, 2 μm thick SiO₂ was deposited by plasma-enhanced chemical vapor deposition (PECVD) to serve as both

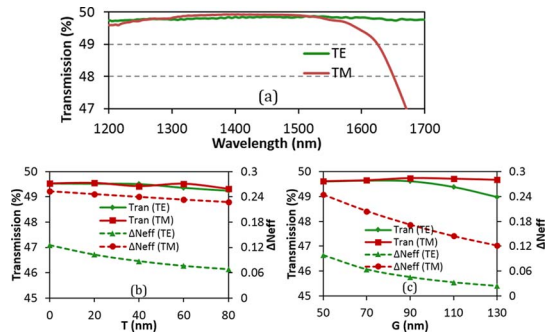


Fig. 3. (a) Simulated transmission spectra (outputs 1 and 2 have identical spectra) of the proposed splitter. (b) Change of transmission efficiency and ΔN_{eff} with T and with (c) G at 1550 nm.

the passivation layer and the etching mask for the Bosch process. The Bosch process was used to deeply etch the silicon substrate so that the chip facet can be accessed by a lensed fiber [18]. After the Bosch process, the top SiO₂ left was measured to be 1.3 μm thick. The scanning electron microscope (SEM) image of the fabricated adiabatic splitter is shown in Fig. 4(b). Based on the SEM measurements, the variation on T and G was around ± 5 nm.

To characterize the splitter performance more reliably, three stages of 1×2 splitters were cascaded to form a 1×8 splitter, as shown in Fig. 4(a). The average EL of a single element was then estimated based on the following equation [19]:

$$\text{EL(dB)} = (P_o - P_{\text{wg}})/3 + 3,$$

where P_o is the average output power of all output ports, and P_{wg} is the measured output power of the reference strip waveguide. The factors 1/3 and 3 correspond to the number of cascaded stages and the 50% splitting ratio, respectively.

The schematic of the measurement setup is shown in Fig. 4(c). An ASE source (THORLABS ASE730) with an operation wavelength of 1530 to 1610 nm was used as the wideband light source, and the output passed through an in-line polarizer and a polarization controller (PC) to generate the single-polarized light. A high-efficiency spot-size converter [18] was used on-chip for the edge couplings between the silicon chip and tapered single-mode fibers. The output spectrum of the splitter was monitored by an optical spectrum analyzer (OSA). Because the operation wavelength of the polarizer used was limited to 1600 nm, only the spectrum within the range of 1530 to 1600 nm was extracted for device performance characterizations, and the results are shown in Fig. 5.

Figures 5(a) and 5(b) show the measured excess loss of the splitter for TE and TM polarizations, respectively. The insertion losses of three random ports [as indicated in Fig. 4(a)] were also extracted to show the uniformity of the 1×8 splitter. It can be seen that a flat spectrum and low-loss profile can be achieved for both polarizations over the entire measured wavelength range. Due to the higher sensitivity to sidewall roughness, the performance of TE-polarized light has larger fluctuations across the operation bandwidth. For better performance characterizations, regression lines were extracted based on the robust locally weighted regression method [20]. It is clear that the excess loss is lower than 0.19 dB over the whole measured wavelength range for the TE polarization and better than 0.14 dB for the TM polarization. Compared with simulation results, the measurement results agree well for the TM polarization, while they are slightly worse for the TE polarization. This is mainly due to the extra scattering losses caused by sidewall roughness of the tiny gaps between tapers where the

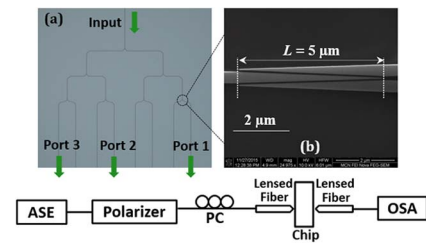


Fig. 4. (a) Optical microscope image of the cascaded 1×8 splitter. (b) Enlarged view (SEM image) of the fabricated taper region. (c) Schematic illustration of the measurement setup.

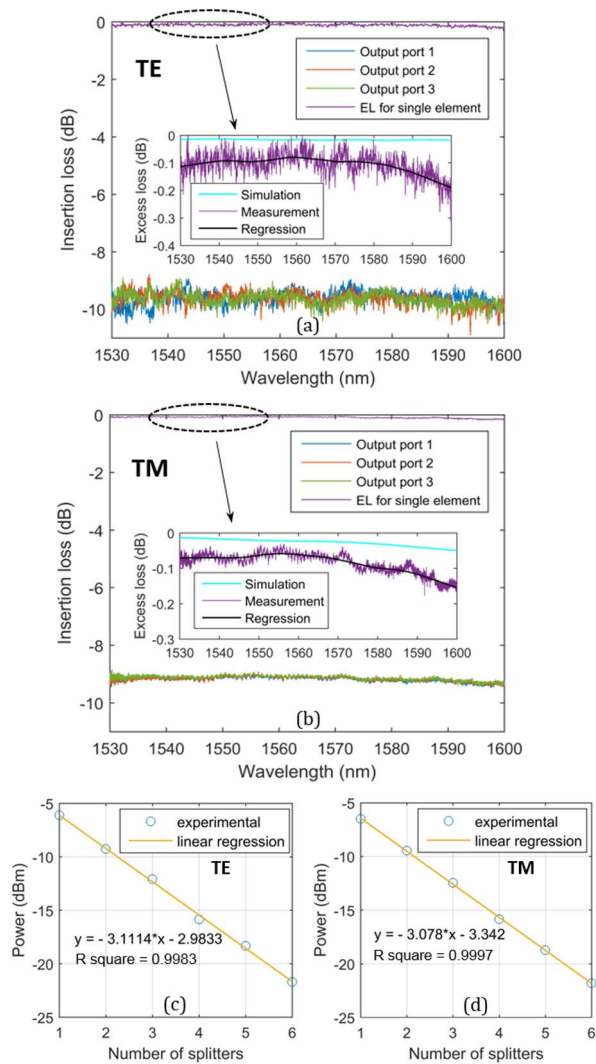


Fig. 5. Experimentally measured spectra of the splitter for (a) the TE and (b) the TM polarization, and the extracted loss at 1550 nm for (c) the TE polarization and (d) the TM polarization. R square is the calculated coefficient of determination.

TE field is mainly concentrated. This can be further improved by optimizing the fabrication processes (e.g., oxidation) to have smoother sidewalls.

More accurate results on the excess loss at 1550 nm were extracted by linearly fitting the measured output power after passing through different numbers of splitters, and the results are shown in Figs. 5(c) and 5(d). At 1550 nm, an ultra-low excess loss of 0.11 and 0.08 dB can be achieved for the TE and the TM polarizations, respectively, and the results also agree well with the measured wideband spectrum shown in Figs. 5(a) and 5(b).

The power uniformity of the splitter was also estimated, and it was defined as the difference between the maximum output power and the minimum output power across all output ports of the splitter. The power uniformity increases with the number of cascade stages [19]. For the cascaded 1×8 splitter, as shown by the insertion loss spectra in Figs. 5(a) and 5(b), uniform outputs are achieved over a wide wavelength range. At 1550 nm, the power uniformity across all eight output ports was measured to be only 0.47 and 0.17 dB for the TE and TM polarizations,

respectively. For a single 1×2 splitter, the best measured uniformity was only 0.07 and 0.02 dB, respectively.

In conclusion, a broadband, low-loss, and polarization insensitive optical power splitter based on adiabatic tapered silicon waveguides has been proposed and experimentally demonstrated. The proposed splitter has been shown to be only 5 μm in length, and the EL is <0.19 dB for the TE polarization and <0.14 dB for the TM polarization over the entire measured wavelength range of 1530 to 1600 nm. For the proposed splitter, it is possible to realize various splitting ratios by using asymmetric gap sizes between tapers, which will be investigated in future research. The relatively small taper tip width and the gap size would make the splitter hard to realize through today's standard CMOS process. However, with the rapid development of the fabrication technology, a much smaller feature size can be realized, such as using the emerging extreme-UV lithography, and the proposed device is expected to have a strong potential in applications that need ultra-high-efficiency optical power distributions and broadband operations.

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